Unit 04: Data Base Management Systems

CONTENTS

Objectives

After studying this unit, you will be able to:

- Explain Database.
- Discuss DBMS and its components.
- Know about the database models
- Understand working with database.
- Explain database at work.
- Explain various common corporate DBMS.

Introduction

Data are units of information, often numeric, that are collected through observation. In a more technical sense, data are a set of values of qualitative or quantitative variables about one or more persons or objects, while a datum (singular of data) is a single value of a single variable. Data is the raw material from which useful information is derived. It is defined as raw facts or observations. Data is a collection of unorganized facts but can be made organized into useful information (**Error! Reference source not found.**). A set of values of qualitative or quantitative variables about one or more persons or objects. Data is employed in scientific research, businesses management (e.g., sales data, revenue, profits, stock price), finance, governance (e.g., crime rates, unemployment rates, literacy rates), human organizational activity (e.g., censuses of the number of homeless people by non-profit organizations).

Data is basically a collection of raw facts and figures we have a rough material that can be possessed by any computing machine that actually pertains to data. Example, we have few numbers with us, but which number it is referring to, it is unknown. We have no idea if it is specifying somebody's age, or it is specifying some currency what it is specifying we don't know. It is just a number. Okay, so actually the collection of facts, which can be in different kind of conclusions can be drawn from it that is actually called as data.

Information pertains to a systematic and meaningful form of data. It is the knowledge acquired through study or experience. The information helps human beings in their decision-making. Information can be said to be data that has been processed in such a way as to increase the knowledge of the person who uses the data.

Knowledge is the understanding based on extensive experience dealing with information on a subject. It is a familiarity, awareness, or understanding of someone or something, such as facts (descriptive knowledge), skills (procedural knowledge), or objects (acquaintance knowledge). By most accounts, knowledge can be acquired in many different ways and from many sources, including but not limited to perception, reason, memory, testimony, scientific inquiry, education, and practice. The philosophical study of knowledge is called epistemology.

Figure 1: Conversion of Raw Data into Information

Let us consider an example of Mount Everest. Most of the books they site out what is the height of Mount Everest, that is just a data. The height can be measured precisely with any altimeter and entered into a particular database. This data can be included in any particular book, along with other data on Mount Everest like how many trees are there or other things related to Mount Everest what is a sea level height. All this data can describe the mountain in a manner that would be useful for those who wish to actually make a decision about whether they should decide out the best manner how they can climb it. Thereby an understanding that is based on experience of climbing the mountains, that could actually be used by some other persons on a way that they can reach the Mount Everest. That is actually the knowledge. However, when we consider that the practical way of actually climbing of Mount Everest by a person climbs, based on his knowledge may be considered as his wisdom. In other words, we can say that wisdom refers to the practical application of a person's ability or a person's knowledge, depending upon the kind of circumstances, which are mostly considered as good.

4.1 Data Processing

The process of converting the facts into meaningful information. Data processing refers to the process of performing specific operations on a set of data or a database. As illustrated in Figure 2, Input one input two input three there are so many different kinds of data processing stages that are involved, like pre-processing is there storage is there, then manipulations are there analysis is there and we have so many different kinds of stages involves sorting is also involved over here, and then an output is generated so this output is the meaningful information or we can say organized data.

Example, every year about one lakh students take admission in National Open School (NOS). If you are asked to memorise records of date of birth, subjects offered and postal address of all these students, it will not be possible for you. In order to deal with such problems, we construct a database where in such situations, all information about students can be arranged in a tabular form. Several records can be maintained.

A database is an organized collection of facts and information, such as records on employees, inventory, customers, and potential customers.

Figure 2: Data Processing Scenario

Metadata: Metadata is the data that describes the properties or characteristics of other data, like we say we have a cat. We have some data about the cat, the name, we can specify the height, we can specify that as a metadata. Therefore, all such data describes the properties or characteristics of other data.

4.2 Database

A database is a system intended to organize, store, and retrieve large amounts of data easily. It consists of an organized collection of data for one or more uses, typically in digital form. One way of classifying databases involves the type of their contents, for example: bibliographic, document-text, statistical. digital databases are managed using database management systems, which store database contents, allowing data creation and maintenance, and search and other access.

A database is a collection of information that is organized so that it can easily be accessed, managed, and updated. In one view, databases can be classified according to types of content: bibliographic, full-text, numeric, and images.

In computing, databases are sometimes classified according to their organizational approach. The most prevalent approach is the relational database, a tabular database in which data is defined so that it can be reorganized and accessed in a number of different ways. A distributed database is one that can be dispersed or replicated among different points in a network. An object-oriented programming database is one that is congruent with the data defined in object classes and subclasses.

Computer databases typically contain aggregations of data records or files, such as sales transactions, product catalogs and inventories, and customer profiles. Typically, a database manager provides users the capabilities of controlling read/write access, specifying report generation, and analyzing usage.

4.3 Types of Databases

There are different types of database as discussed below:

Analytical Database: Analysts may do their work directly against a data warehouse or create a separate analytic database for Online Analytical Processing (OLAP). For example, a company might extract sales records for analyzing the effectiveness of advertising and other sales promotions at an aggregate level.

Data Warehouse: Data warehouses archive modern data from operational databases and often from external sources such as market research firms. Often operational data undergoes transformation on its way into the warehouse, getting summarized, anonym zed, reclassified, etc. The warehouse becomes the central source of data for use by managers and other end-users who may not have access to operational data.

Distributed Database: These are databases of local work-groups and departments at regional offices, branch offices, manufacturing plants and other work sites. These databases can include segments of both common operational and common user databases, as well as data generated and used only at a user's own site.

End User Database: These databases consist of data developed by individual end-users. Examples of these are collections of documents in spread sheets, word processing and downloaded files, even managing their personal baseball card collection.

External Log Database: These databases contain data collected for use across multiple organizations, either freely or via subscription. The Internet movie database is one example.

Hypermedia Log Database: World Wide Web can be thought of as a database, spread across millions of independent computing systems. Web browsers "process" this data one page at a time, while Web crawlers and other software provide the equivalent of database indexes to support search.

Operational Log Database: These databases store detailed data about the operations of an organization. They are typically organized by subject matter, process relatively high volumes of updates using transactions. Essentially every major organization on earth uses such databases. Examples include customer databases that record contact, credit, and demographic information about a business' customers, personnel databases that hold information such as salary, benefits, and skills data about employees, Enterprise resource planning that record details about product components, parts inventory, and financial databases that keep track of the organization's money, accounting and financial dealings.

Customer Database: Such databases record contact, credit, and demographic information about a business' customers.

Personnel Database: It can hold information such as salary, benefits, skills data about employees.

Enterprise resource planning: Such database records details about product components, parts inventory.

Financial Database: Such database tracks an organization's money, accounting and financial dealings.

Purpose of Using a Database

A database is typically designed so that it is easy to store and access information. A good *database is crucial* to any company or organization. This is because the **database** stores all the important details about the company such as employee records, transactional records, salary details. Database management systems are essential for businesses because they offer an efficient way of handling large amounts and multiple types of data. The ability to access data efficiently allows companies to make informed decisions quicker.

4.4 Database Administrator (DBA)

A Database Administrator (DBA) in Database Management System (DBMS) is an IT professional who works on creating, maintaining, querying, and tuning the database of the organization. They are also responsible for maintaining data security and integrity.

Database Manager provides users the capabilities of controlling read/write access, specifying report generation, and analyzing usage. They are prevalent in large mainframe systems, but are also present in smaller distributed workstation and mid-range systems such as the AS/400 and on personal computers. A database manager (DB manager) is a computer program, or a set of computer programs, that provide basic database management functionalities including creation and maintenance of databases. Database managers have several capabilities including the ability to back up and restore, attach and detach, create, clone, delete and rename the databases.

Database Three Level Architecture used for describing the structure of specific database systems. In this architecture, the database schemas can be defined at three levels (Figure 3).

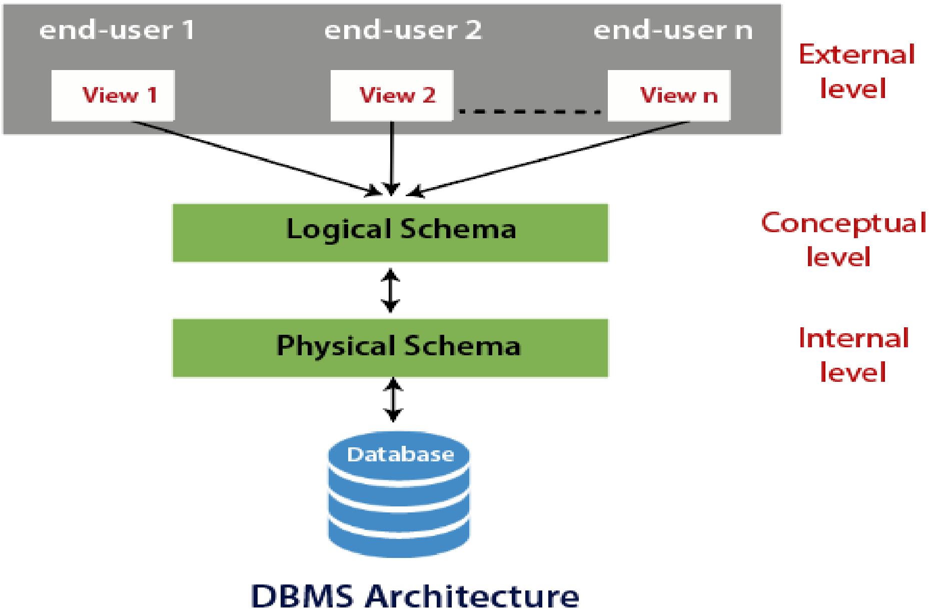

Figure 3: Database- Three Level Architecture

- **Internal/physical level:** Shows how data are stored inside the system. It is the closest level to the physical storage. This level is also known as physical level. This level describes how the data is actually stored in the storage devices. This level is also responsible for allocating space to the data. This is the lowest level of the architecture.

  The physical level keeps the information about the actual representation of the entire database, that is, the actual storage of the data on the disk in the form of records or blocks.
- **Conceptual/logical level:** It is also called **logical level**. The whole design of the database such as relationship among data, schema of data etc. are described in this level. Database constraints and security are also implemented in this level of architecture. This level is maintained by DBA (database administrator).

  This level lies in between the user level and the physical storage view. Only one conceptual view for single database us available. It hides the details of physical storage structures and concentrates on describing entities, data types, relationships, user operations, and constraints.
- **External level:** It is also called **view level**. The reason this level is called "view" is because several users can view their desired data from this level which is internally fetched from database with the help of conceptual and internal level mapping. This is the highest or top level of data abstraction (No knowledge of DBMS Software and Hardware or physical storage). This level is concerned with the user. Each external schema describes the part of the database that a particular user is interested in and hides the rest of the database from user. There can be n number of external views for database where 'n' is the number of users. Example: A accounts department may only be interested in the student fee details. All the database users work on external level of DBMS.

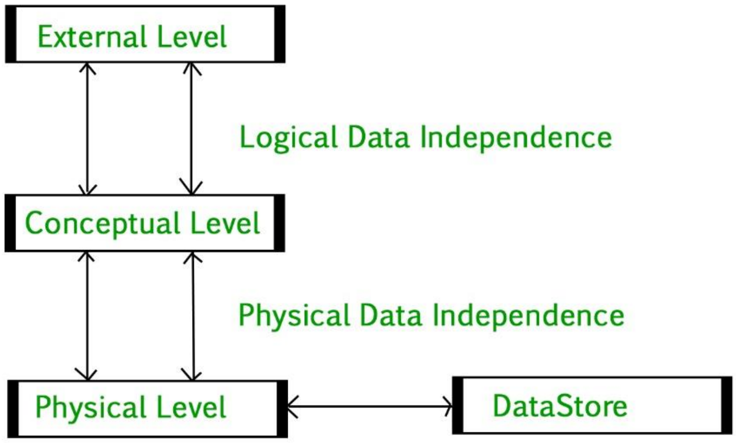

Figure 4: Data Independence

Data Independence

Data Independence is defined as a property of DBMS that helps you to change the database schema at one level of a database system without requiring changing the schema at the next higher level (Figure 4). Data independence helps you to keep data separated from all programs that make use of it. There are two types of data independence:

Physical Data Independence can be defined as the capacity to change the internal schema without having to change the conceptual schema. If we do any changes in the storage size of the database system server, then the conceptual structure of the database will not be affected. It is used to separate conceptual levels from the internal levels. It occurs at the logical interface level.

Logical Data Independence refers characteristic of being able to change the conceptual schema without having to change the external schema. It is used to separate the external level from the conceptual view. If we do any changes in the conceptual view of the data, then the user view of the data would not be affected. It occurs at the user interface level.

Database System Applications

Banking: The banks can use database for storing the customer information, their account details loans that have been sanctioned out, and the different kinds of banking transactions can be also listed out. Thereby, database can be used for storing the customer information, accounts, and loans, and banking transactions.

Airlines: The aviation sector can use database as a basis for reservations and for scheduling different kinds of information.

Universities: A university can maintain the student information and also can maintain the employee database. It can consider out the grade achieved by the students; it can consider out the registrations that are done for a particular course over there.

Credit card Transactions: For purchases on credit cards and generation of monthly statements.

Telecommunication: For keeping records of calls made, generating monthly bills, maintaining balances on prepaid calling cards, and storing information about the communication networks.

Finance: For storing information about holdings, sales, and purchases of financial instruments such as stocks and bonds.

Sales: For customer, product, & purchase information.

Manufacturing: For management of supply chain & for tracking production of items in factories, inventories of items in warehouses/ stores, and orders for items.

Human Resources: For information about employees, salaries, payroll taxes and benefits, and for the generation of paychecks. Mahindra and Mahindra organization are collecting the data from DTO and feeding into database for the purpose of decision making.

4.5 Database Management Systems

The DBMS is a collection of interrelated data and a set of programs to access those data. It can be said to be a software that interacts with end users, applications, and the database itself to capture and analyze the data. It encompasses the core facilities provided to administer the database. The sum total of the database, the DBMS and the associated applications can be referred to as a "database system". The term "database" is often used to loosely refer to any of the DBMS, the database system or an application associated with the database. The data in the databases is viewed as rows and columns in a series of tables, and the vast majority use Structured Query Language (SQL) for writing and querying data.

Purpose of DBMS

- To avoid data redundancy
- To avoid data inconsistency
- Data independence
- Efficient data access
- Data integrity and security
- Data administration
- Concurrent access and crash recovery
- Reduced application development time

Components of DBMS

DBMS consist of five major components (Figure 5):

End Users: All the modern applications, web or mobile, store user data. Applications are programmed in such a way that they collect user data and store the data on DBMS systems running on their server. The users can store, retrieve, update and delete data.

Software: The main component or program that controls everything. It is more like a wrapper around the physical database. It provides an easy-to-use interface to store, access and update data. It capable of understanding the Database Access Language and interpret into actual database commands to execute them on DB.

Hardware: Computer, hard disks, I/O channels for data, Keyboard with which we type commands, computer's RAM, ROM and any other physical component involved before any data is successfully stored into the memory.

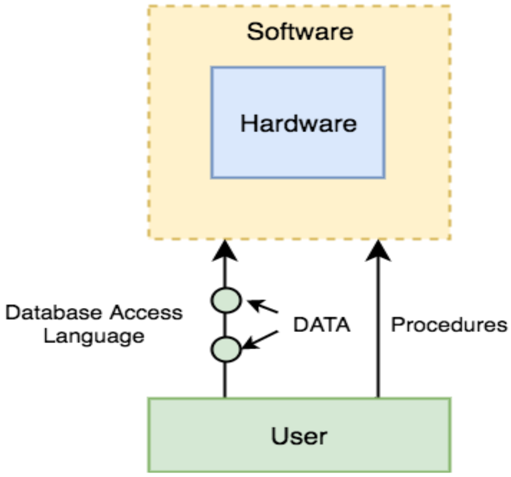

Figure 5: Components of DBMS

Data: Data is the resource, for which DBMS was designed. Primary motive is creation of DBMS was to store and utilize data.

Procedures: Procedures pertain to the general instructions to use a DBMS. It includes instructions to setup and install a DBMS, to login and logout of DBMS software, to manage databases, to take backups, generating reports etc.

Database Access Language: A simple language designed to write commands to access, insert, update and delete data stored in any database. The users can write commands in the Database Access Language and submit it to the DBMS for execution, which is then translated and executed by the DBMS. The users can create new databases, tables, insert data, fetch stored data, update data & delete data.

4.6 Database Models

Data model help to represent data and to make the data understandable. A modeling language is a data modeling language to define the schema of each database hosted in the DBMS, according to the DBMS database model. DBMS are designed to use one of five database structures to provide simplistic access to information stored in databases. The five database structures are:

- Object model
- Hierarchical model
- Network model
- Relational model
- Multidimensional model

Levels of Data Models

Object-based and record-based data models are used to describe data at the conceptual and external levels, the physical data model is used to describe data at the internal level. The object-based data models use concepts such as entities, attributes, and relationships. There are two common types: The *entity-relationship model* has emerged as one of the main techniques for modeling database design and forms the basis for the database design methodology. The *object-oriented data model* extends the definition of an entity to include, not only the attributes that describe the state of the object but also the actions that are associated with the object, that is, its behavior.

Entities are represented by means of rectangles. Rectangles are named with the entity set they represent. As illustrated, we have three entities over here student, teacher and projects (Figure 6).

Figure 6: Entities in ER-Model

Attributes are the properties of entities. These are represented by means of ellipses. Every ellipse represents one attribute and is directly connected to its entity (rectangle) (Figure 7).

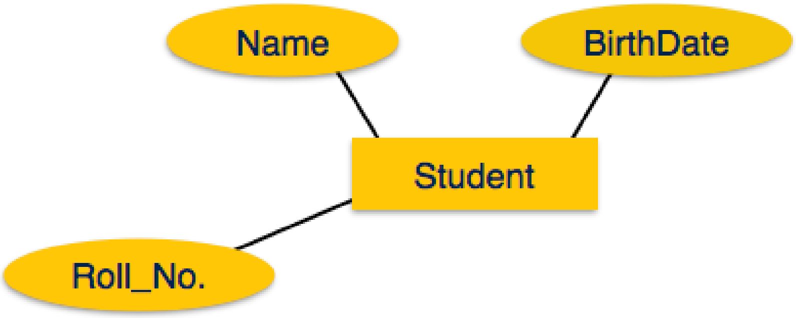

Figure 7: Entities and Attributes in ER-Model

Composite Attributes: If the attributes are composite, they are further divided in a tree like structure. This is a kind of attribute that are composite in nature. They can be further divided out into tree like structure, like a name, name is an attribute. It can consist of two further attributes, that is why we call them as composite attributes. Example, as illustrated in Figure 8, the name can further be segregated into first name and the last name which are composite attributes.

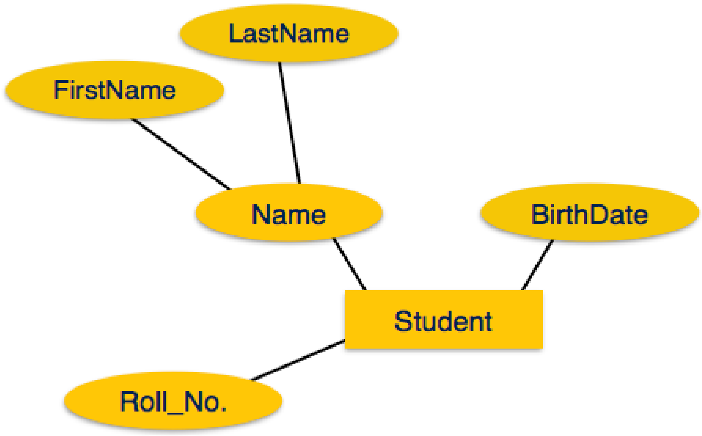

Figure 8: Composite Attributes

Multivalued Attributes: Multivalued attributes are depicted by double ellipse like phone number (Figure 9). It can have multiple values such as a student can have multiple phone numbers. We call it as multivalued attribute as it cannot be a unique to a particular student, a student can have multiple phone numbers as well.

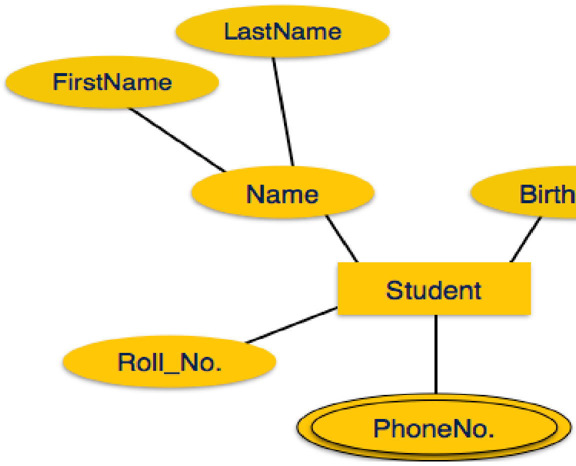

Figure 9: Multivalued Attribute

Derived Attributes: Derived attributes are depicted by dashed ellipse. In Figure 10, age is a derived attribute. Every student is having some age. Age is dependent upon a student. Like birth date is one attribute of a student entity. The age can only be derived from after knowing the birthdate of a particular students so that is why we call it as derived attribute.

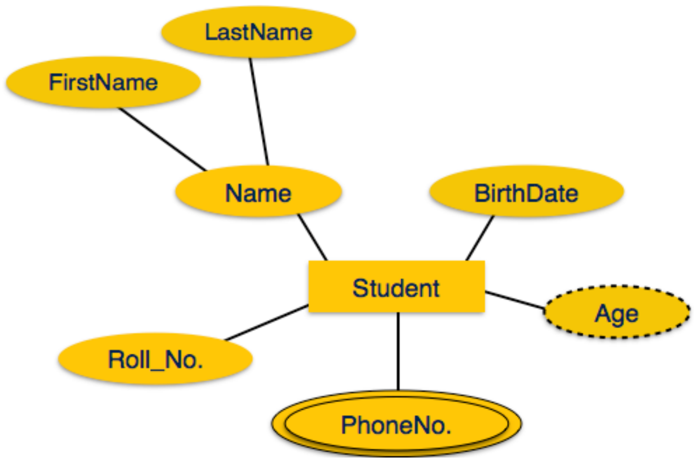

Figure 10: Derived Attribute

Relationships in the ER-model are represented by diamond-shaped box. The name of the relationship is written inside the diamond-box. All the entities (rectangles) participating in a relationship, are connected to it by a line. Figure 11 depicts the sample entity-relationship diagram.

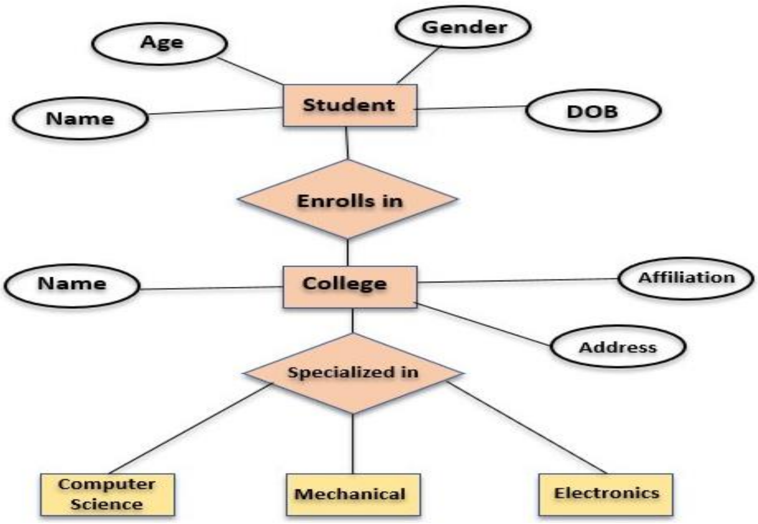

Figure 11: Sample ER-Diagram

Hierarchical database model is the oldest database models. It is like a structure of a tree with the records forming the nodes and fields forming the branches of the tree. The records are forming the nodes, and the fields are forming the branches of a particular tree. As illustrated in Figure 12, we have a root, that is the node (the record). Then, we have the subfields over there that are actually forming the branches of a particular tree that is child with further sub-levels. There are different levels that are involved in a hierarchical database model.

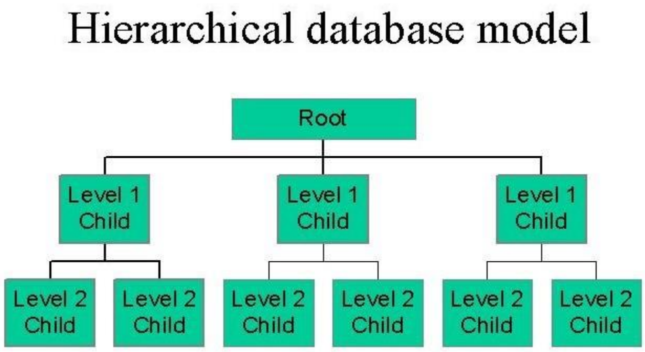

Figure 12: Hierarchical Database Model

Network Model replaces the hierarchical tree with a graph thus allowing more general connections among the nodes (Figure 13). The main difference of the network model from the hierarchical model, is its ability to handle many to many (N:N) relations.

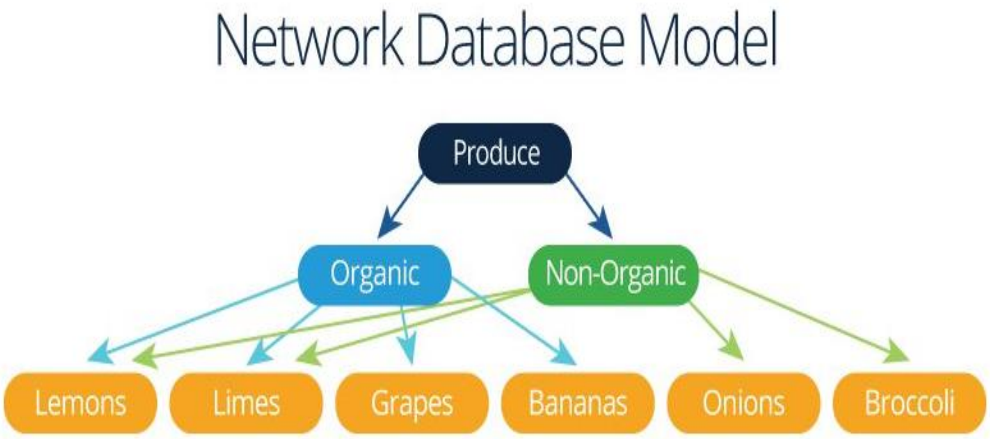

Figure 13: Network Database Model

Relational Model is the most commonly used today. It is used by mainframe, midrange and microcomputer systems. It uses two-dimensional rows and columns to store data (Figure 14). The tables of records can be connected by common key values. While working for IBM, E.F. Codd designed this structure in 1970. The model is not easy for the end user to run queries with because it may require a complex combination of many tables.

Advantages of Relational Model

- Simple: Relational model is simpler as compared to the network and hierarchical model.
- Scalable: It can be easily scaled as we can add as many rows and columns we want.
- Structural Independence: We can make changes in database structure without changing the way to access the data

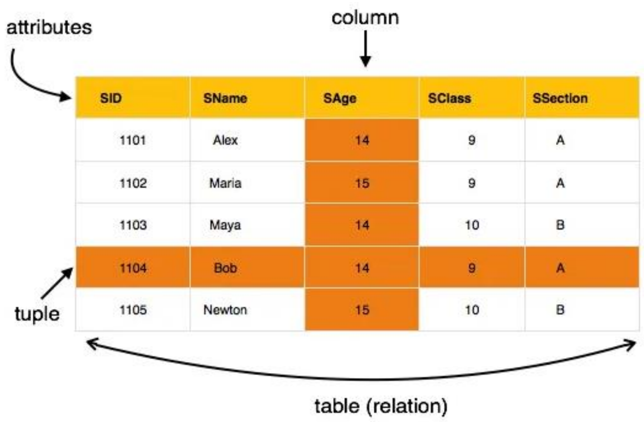

Figure 14: Depiction of Relational Database Model

Multidimensional Model is similar to the relational model. The dimensions of the cube-like model have data relating to elements in each cell. This structure gives a spread sheet-like view of data. This structure is easy to maintain because records are stored as fundamental attributes— in the same way they are viewed— and the structure is easy to understand. Its high performance has made it the most popular database structure when it comes to enabling OLAP.

4.7 Working with Database

A database is an organized collection of data, generally stored and accessed electronically from a computer system. Where databases are more complex, they are often developed using formal design and modelling techniques.

Structured Query Language (SQL)

Structured Query Language (SQL) is the database language. It helps to perform certain operations on the existing database and also can be used to create a database. It uses certain commands like Create, Drop, Insert etc. to carry out the required tasks. There are different data types in SQL such as:

- char(n): Fixed length character string, with user-specified length *n.*
- varchar(n): Variable length character strings, with user-specified maximum length *n.*
- Int: Integer (a finite subset of the integers or numbers that is machine-dependent).

Types of SQL Commands

Data Definition Language (DDL): These are used to change the structure of a database and database objects. These can be used to add, remove, or modify tables with in a database (Table 1). There are various DDL commands such as CREATE, ALTER, DROP, TRUNCATE, RENAME etc.

CREATE: Used to create table in the relational database. This can be done by specifying the names and datatypes of various columns. The syntax:

CREATE TABLE *table_name*

```
(
  column1 datatype,
  column2 datatype,
  column3 datatype, ....
);
```

ALTER: Alter command is used for altering the table in many forms like: add a column; rename existing column; drop a column; modify the size of the column or change data type of the column. Example: If a user wants to add a column in a table, following syntax is used:

ALTER TABLE *table_name*
ADD *column_name datatype*;

Data Manipulation Language (DML): SQL commands that deal with the manipulation of data present in the database belong to DML or Data Manipulation Language. These include most of the SQL statements. They are:

- o INSERT- Used to insert data into a table.
- o UPDATE– Used to update existing data within a table.
- o DELETE- Used to delete records from a database table.

INSERT: The INSERT statement is a SQL query used to insert data into the row of a table. The syntax for it is:

INSERT INTO TABLE_NAME (col1, col2, col3, ..., col N)

VALUES (value1, value2, value3, ... valueN);

Or

INSERT INTO TABLE_NAME

VALUES (value1, value2, value3, ... valueN);

UPDATE: This command is used to update or modify the value of a column in the table. The syntax is:

UPDATE table_name SET [column_name1= value1,...column_nameN = valueN]

[WHERE CONDITION]

DELETE: It is used to remove one or more row from a table. The syntax is:

DELETE FROM table_name [WHERE condition];

Table 1: Comparison of DDL with DML

S. No.	DDL	DML
1.	Stands for Data Definition Language	Stands for Data Manipulation Language
2.	Used to create database schema and can be used to define some constraints as well.	Used to add, retrieve or update the data.
3.	Basically defines the column (Attributes) of the table.	Used to add or update the row of the table. These rows are called as tuple.
4.	Doesn't have any further classification.	Further classified into Procedural and Non-Procedural DML.

Data Control Language (DCL): These commands are used to grant and take back authority from any database user. Some commands that come under DCL are:

- o Grant: It is used to give user access privileges to a database. The syntax is:

  GRANT SELECT, UPDATE ON MY_TABLE TO SOME_USER, ANOTHER_USER;

- o Revoke: It is used to take back permissions from the user. The syntax is:

  REVOKE SELECT, UPDATE ON MY_TABLE FROM USER1, USER2;

Data Query Language (DQL): Such commands are used to fetch the data from the database. It uses only one command:

SELECT: This is the same as the projection operation of relational algebra. It is used to select the attribute based on the condition described by WHERE clause. SELECT statement is used to select data from a database. The data returned is stored in a result table, called the result-set. The syntax for SELECT command is:

```
SELECT expressions
FROM TABLES
WHERE conditions;
```

For example:

```
SELECT emp_name
FROM employee
WHERE age > 20;
```

4.8 Databases at Work

Databases have been a staple of business computing from the very beginning of the digital era. In fact, the relational database was born in 1970. Originally, databases were flat, means, that the information was stored in one long text file, called a tab delimited file. Each entry in the tab delimited file is separated by a special character, such as a vertical bar (|). Each entry contains multiple pieces of information (fields) about a particular object or person grouped together as a record. The text file makes it difficult to search for specific information or to create reports that include only certain fields from each record. One has to search sequentially through the entire file to gather related information, such as age or salary.

Relational databases have grown in popularity to become the standard. Unlike traditional databases, the relational database allows you to easily find specific information. It allows sorting based on any field and generate reports that contain only certain fields from each record. Relational databases use tables to store information. Relational database model takes advantage of this uniformity to build completely new tables out of required information from existing tables. Each table contains a column or columns that other tables can key on to gather information from that table. A typical large database, like the one a big Web site, such as Amazon would have, will contain hundreds or thousands of tables, all used together to quickly find the exact information needed at any given time.

The major uses and characteristics of relational database are:

- One can quickly compare salaries and ages because of the arrangement of data in columns.
- Uses the relationship of similar data to increase the speed and versatility of the database.
- The "relational" part of the name comes into play because of mathematical relations.
- A typical relational database has anywhere from 10 to more than 1,000 tables.
- Relational databases are created using a special computer language, structured query language (SQL), that is the standard for database interoperability.
- SQL is the foundation for all of the popular database applications available today, from Access to Oracle.

Database Transaction comprises of a unit of work performed within a DBMS (or similar system) against a database, and treated in a coherent and reliable way independent of other transactions. The transactions in a database environment have two main purposes:

- To provide reliable units of work that allow correct recovery from failures and keep a database consistent even in cases of system failure, when execution stops (completely or partially) and many operations upon a database remain uncompleted, with unclear status.
- To provide isolation between programs accessing a database concurrently.

The database transactions are governed under few considerations such as:

A database transaction must be atomic, consistent, isolated and durable. Such properties of database transactions are cited by the database practitioners using the acronym ACID. Further, the system must isolate each transaction from other transactions, results must conform to existing constraints in the database, and transactions that complete successfully must get written to durable storage.

4.9 Common Corporate Database Management Systems

Different types of software applications have been used in the past and may be still in use on older, legacy systems at various organizations around the world. However, these examples provide an overview of the most popular and most-widely used by IT departments. Some typical examples of DBMS include: Oracle, IBM DB2, Microsoft Access, Microsoft SQL Server, PostgreSQL, MySQL, and FileMaker.

ORACLE

Oracle Database (commonly referred to as Oracle RDBMS or simply as Oracle) is an object-relational database management system (ORDBMS) produced and marketed by Oracle Corporation. Larry Ellison and his friends and former co-workers Bob Miner and Ed Oates started the consultancy Software Development Laboratories (SDL) in 1977. SDL developed the original version of the Oracle software. The name Oracle comes from the code-name of a CIA-funded project Ellison had worked on while previously employed by Ampex.

IBM DB2

IBM DB2 Enterprise Server Edition is a relational model database server developed by IBM. It primarily runs on UNIX (namely AIX), Linux, IBM i (formerly OS/400), z/OS and Windows servers. DB2 also powers the different IBM InfoSphere Warehouse editions. Alongside DB2 is another RDBMS: Informix, which was acquired by IBM in 2001.

Microsoft Access

Microsoft Office Access, previously known as Microsoft Access, is a relational database management system from Microsoft that combines the relational Microsoft Jet Database Engine with a graphical user interface and software-development tools. It is a member of the Microsoft Office suite of applications, included in the Professional and higher editions or sold separately.

Access stores data in its own format based on the Access Jet Database Engine. It can also import or link directly to data stored in other applications and databases.

Software developers and data architects can use Microsoft Access to develop application software, and "power users" can use it to build simple applications. Like other Office applications, Access is supported by Visual Basic for Applications, an object-oriented programming language that can reference a variety of objects including DAO (Data Access Objects), ActiveX Data Objects, and many other ActiveX components. Visual objects used in forms and reports expose their methods and properties in the VBA programming environment, and VBA code modules may declare and call Windows operating-system functions.

Microsoft SQL Server

Microsoft SQL Server is a relational model database server produced by Microsoft. Its primary query languages are T-SQL and ANSI SQL. Since parting ways, several revisions have been done independently. SQL Server 7.0 was a rewrite from the legacy Sybase code. It was succeeded by SQL Server 2000, which was the first edition to be launched in a variant for the IA-64 architecture.

PostgreSQL

PostgreSQL, often simply Postgres, is an object-relational database management system (ORDBMS). It is released under an MIT-style license and is thus free and open source software. The name refers to the project's origins as a "post-Ingres" database, the original authors having also developed the Ingres database. (The name Ingres was an abbreviation for INteractive Graphics REtrieval System.). However, the PostgreSQL Core Team announced in 2007 that the product would continue to use the name PostgreSQL. The name refers to the project's origins as a "post-Ingres" database, the original authors having also developed the Ingres database.

MySQL

MySQL is a relational database management system (RDBMS) that runs as a server providing multi-user access to a number of databases. It is named after developer Michael Widenius' daughter, 'My'. The SQL phrase stands for Structured Query Language. The MySQL development project has made its source code available under the terms of the GNU General Public License, as well as under a variety of proprietary agreements. MySQL was owned and sponsored by a single for-profit firm, the Swedish company MySQL AB, now owned by Oracle Corporation.

For commercial use, several paid editions are available, and offer additional functionality. Some free software project examples: Joomla, WordPress, Mob, phpBB, Drupal and other software built on the LAMP software stack. MySQL is also used in many high-profile, large-scale World Wide Web products, including Wikipedia, Google (though not for searches) and Face book.

FileMaker

FileMaker Pro is a cross-platform relational database application from FileMaker Inc., formerly Claris, a subsidiary of Apple Inc. It integrates a database engine with a GUI-based interface, allowing users to modify the database by dragging new elements into layouts, screens, or forms. The current versions are: FileMaker Pro 11, FileMaker Pro 11 Advanced, FileMaker Pro 11 Server, FileMaker Pro 11 Server Advanced and FileMaker Go.

FileMaker evolved from a DOS application, but was then developed primarily for the Apple Macintosh. Since 1992 it has been available for Microsoft Windows as well as Mac OS, and can be used in a heterogeneous environment. It is available in desktop, server, iOS and web-delivery configurations.

Summary

- Database is a system intended to organize, store and retrieve large amounts of data easily.
- DBMS is a tool that is employed in the broad practice of managing databases.
- Distributed database management system (ODBMS) is collection of data which logically belong to the same system but are spread out over the sites of the computer network.
- A modelling language is a data modelling language to define the scheme of each database hosted in DBMS.
- The end-user databases consist of data developed by individual end-users.
- Data structures optimized to deal with very large amount of data stored on a permanent data storage device.
- The operational databases store detailed data about the operations of an organization.

Keywords

Analytical database: Analysts may do their work directly against a data warehouse or create a separate analytic database for Online Analytical Processing (OLAP).

Data definition subsystem: It helps the user create and maintain the data dictionary and define the structure of the files in a database.

Data structure: Data structures optimized to deal with very large amounts of data stored on a permanent data storage device.

Data warehouse: Data warehouses archive modern data from operational databases and often from external sources such as market research firms.

Database: A database is a system intended to organize, store, and retrieve large amounts of data easily. It consists of an organized collection of data for one or more uses, typically in digital form.

Distributed database: These are databases of local work-groups and departments at regional offices, branch offices, manufacturing plants and other work sites.

Hypermedia databases: WWW can be thought of as a database, albeit one spread across millions of independent computing systems.

Microsoft access: It is a relational database management system from Microsoft that combines the relational Microsoft Jet Database ls.

Modeling language: A modeling language is a data modeling language to define the schema of each database hosted in the DBMS, according to the DBMS database model.

Object database models: In recent years, the object-oriented paradigm has been applied in areas such as engineering and spatial databases, telecommunications and in various scientific domains.

Post-relational database models: The products offering a more general data model than the relational model are sometimes classified as post-relational.

Self Assessment

1. DBMS helps achieve

A. Data independence

B. Centralized control of data

C. Neither A nor B

D. Both A and B

2. Relational model stores data in the form of_________.

A. Maps

B. Pages

C. Tables

D. files

3. ………………… is a full form of SQL.

A. Standard query language

B. Sequential query language

C. Structured query language

D. Server-side query language

4. Integrity rules that define the procedure to protect the data pertain to ________

A. Integrity

B. Integration

C. Standardization

D. Schema

5. A relational database developer refers to a record as

A. a criteria

B. a relation

C. an attribute

D. a tuple

6. Multidimensional model is useful in

A. CRM systems

B. Online Analytical Processing (OLAP)

C. ERP systems

D. Banking systems

7. Which of the following is not a valid SQL type?

A. FLOAT

B. NUMERIC

C. CHARACTER

D. DECIMAL

8. _____________ database structure is the most commonly used today.

A. Relational structure
B. Network structure
C. Object structure
D. Hierarchical structure

9. ____________ was used in early mainframe DBMS records.
A. Relational structure
B. Network structure
C. Object structure
D. Hierarchical structure

10. The multidimensional structure is similar to the relational model.
A. True
B. False

11. DBMS is a computer hardware program.
A. True
B. False

12. The network structure consists of more
A. Complex relationships
B. Single relationship
C. All of these
D. Double relationship

13. The relational database was born in
A. 1978
B. 1975
C. 1970
D. 1960

14. Which of the following is not a DDL command?
A. TRUNCATE
B. ALTER
C. CREATE
D. UPDATE

15. Which command is used to change the definition of a table in SQL?
A. ALTER
B. CREATE
C. UPDATE
D. SELECT

Answers for Self Assessment

1.	D	2.	C	3.	C	4.	A	5.	D
6.	B	7.	D	8.	A	9.	D	10.	A
11.	B	12.	A	13.	C	14.	D	15.	A

Review Questions

1. What is Database? What are the different types of database?
2. What are analytical and operational database? What are other types of database?
3. Define the Data Definition Subsystem.
4. What is Microsoft Access? Discuss the most commonly used corporate databases.
5. Write the full form of DBMS. Elaborate the working of DBMS and its components?
6. Discuss in detail the Entity-Relationship model?
7. Describe working with Database.
8. What is Object database models? How it differs from other database models?
9. Discuss the data independence and its types?
10. What are the various database models? Compare.
11. Describe the common corporate DBMS?

Further Readings

Introduction to Database Management Systems by Atul Kahate, Pearson Publishers

Web Links

Fundamentals of Database Systems Seventh Edition (iran-lms.com)